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AVERAGES

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Introduction

There are two sections A and B of a class, consisting of 36 and 44 students respectively. If the average weight of section A is 40kg and that of section B is 35kg, find the average weight of the whole class.

Concept and Definitions

Average = Sum Of Observations / No Of Observations

Group 1

Average = A1

Sum of Obs = S1

Number of Obs. = N1

Group 2

Average = A2

Sum of Obs = S2

Number of Obs. = N2

If Group 1 and Group 2 come together then the combined average of the group will be A = S1 + S2 / N1 + N2

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Key Formulae

Problem of Replacement (Most Common Type of Problem in Averages) :

If Average of a group of N people increases/decreases by P kg when one or more persons weighing W kg is replaced by a new person. Then, **the weight of new person is: W+ P x N (if average increases) W- P x N (if average decreases)**

Eg.

The average weight of 5 persons is decreased by 2kg when one of the men whose weight is 50kg is replaced by a new man. The weight of the new man is

In this problem average decrease hence weight of new person is:

50 – 5x2= 40 kg

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Questions

Ques.1 **There are two sections A and B of a class, consisting of 36 and 44 students' respectively. If the average weight of section A is 40kg and that of section B is 35kg, find the average weight of the whole class.**

a) 30 kg

b) 35 kg

c) 37.25 kg

d) 42.50 kg

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Questions

Ques.2 The captain of a cricket team of 11 members is 26 years old and the wicket keeper is 33 years older. If the ages of these two are excluded, the average age of the remaining players is one year less than the average age of the whole team. Find out the average age of the team.

a) 21 years

b) 23 years

c) 24 years

d) 20 years

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Questions

Ques.3 **The average score of a cricketer for ten matches is 38.9 runs. If the average for the first six matches is 42, then find the average for the last four matches.**

a) 33.25

b) 33.5

c) 34.25

d) 35

Questions

- **Ques.4 The distance between two stations A and B is 778 km. A train covers the journey from A to B at 84 km per hour and returns back to A with a uniform speed of 56 km per hour. Find the average speed of train during the whole journey.**

a) 60.3 kmph

b) 35 kmph

c) 57.5 kmph

d) None of these

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Questions

Ques.5 **In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs?**

a) 6.25

b) 5.5

c) 7.4

d) 5

Questions

Ques.6 **The mean weight of a group of seven boys is 56 kg. The individual weights (in kg) of six of them are 52, 57, 55, 60, 59 and 55. Find the weight of the seventh boy.**

a) 64

b) 54

c) 88

d) 28

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Questions

Ques.7 Average of 16 numbers is 8. If 2 is added to every number, the new average will be:

a) 13

b) 9

c) 10

d) 8

Questions

Ques.8 **The average weight of 15 girls in a group is 24 kg when a new girl included the average weight increases by 2. What is the weight of the new girl?**

a) 56 kg

b) 28 kg

c) 34 kg

d) None of these

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Questions

Ques.9 **Average weight of 25 boys in a class is 48 kgs. The average weight of the class of 40 students is 45 kgs. What is the average weight of the 15 girls in the class?**

a) 44 kg

b) 42 kg

c) 40 kg

d) 39 kg

Questions

Ques.10 **The average weight of 8 person's increases by 2.5 kg when a new person comes in place of one of them weighing 65 kg. What might be the weight of the new person?**

a) 76 kg

b) 76.5 kg

c) 85 kg

d) Data inadequate

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Questions

Ques.11 **The average marks of 13 papers is 40. The average marks of the first 7 papers are 42 and that of the last seven papers is 35. Find the marks obtained in the 7th paper** **[Infosys 2025]**

a) 23

b) 38

c) 19

d) 57

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Questions

Ques.12 The average salary of 30 officers in a firm is Rs.120 and the average salary of laborers is Rs. 40. Find the total number of laborers if the average salary of the firm is Rs. 50.

a) 180
b) 420
c) 240
d) 210

Questions

QUES.13 **In 40 overs game, in first 20 overs of a game of cricket, the run rate was only 5. What should be the run rate for the remaining overs so that the total score reaches 300?**

a) 15
b) 10
c) 28
d) 20

Questions

QUES.14 The average of runs scored by a cricketer in five innings is 13 1 /3% more than the average of runs scored by the cricketer in first four innings. If in fifth innings, he makes a century, then find the average of runs scored by him in five innings.

a) 65
b) 58
c) 55
d) None of these

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Questions

QUE.15 The average score (runs/match) of Rahul before the start of World Cup was 35. In the world cup of 6 matches, his total score was 355 runs. Find the total number of matches played by him till date, if his new average after the series is 37.5.

a) 55
b) 56
c) 52
d) 58

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Thank You